[redacted]

CLIENT *CUSTOMER*		LIEU D'INSTALLATION *PLANT LOCATION*	
FONCTION *SERVICE*	**RECYCLED WATER PUMP (10 KG/CM2G)**	QUANTITE NECESSAIRE N° OF UNITS	**4**
ENTRAINE PAR *DRIVEN BY*	**ELECTRIC MOTOR**	SERVICE *OPERATION*	■ CONTINU / *CONTINUOUS* ☐ DISCONTINU / *DISCONTINUOUS*
NOM DU PROJET *PROJECT NAME*		TYPE *TYPE*	**Centrifugal Pump**
REFERENCE OFFRE CONSTRUCTEUR *MFR PROPOSAL N°*			

REV

CONDITIONS OPERATOIRES / *OPERATING CONDITIONS*

LIQUID *LIQUID*	**RECYCLED WATER**	DEBIT Nominal *FLOW Nominal*	**100**	DEBIT Maxi. *FLOW Maxi.*	**120**	m3/h	
		PRESSION REFOULEMENT (1)(2) *DISCHARGE PRESSURE (1)(2)*			**11.4 (5)**	kg/cm²g	
~~PRESSION~~/TEMP. DE REF. ~~*PRESSURE*~~*/TEMP. REF*	**65 °C (5)**	PRESSION ASPIRATION (1)(2) *SUCTION PRESSURE (1)(2)*			**-0.10**	kg/cm²g	
MASSE VOL. A PT DENSITY *AT PT*	**981 kg/m3**	PRESS. DIFFERENTIELLE (1)(2) *DIFFERENTIAL PRESS. (1)(2)*			**11.5**	kg/cm²	
TENSION DE VAPEUR A T VAPOR PRESSURE AT T	**0.25 kg/cm²a**	HAUTEUR MANO. (1)(2) *DIFFERENTIAL HEAD (1)(2)*			**117**	M. C. L.	
VISCOSITE A PT *VISCOSITY AT PT*	**0.4 cP**	NPSH DISPONIBLE (1)(2) *NPSH AVAILABLE (1)(2)*			**7**	M. C. L.	
CORROSION/EROSION DUE A *CORROSION/EROSION CAUSED BY*			**-**				

CARACTERISTIQUES / *PERFORMANCES*

COURBE DE PROPOSITION N° *PROPOSAL CURVE N°*				
NPSH REQUIS (EAU) *REQ'D NPSH (WATER)*			m	
NBRE D'ETAGES *No OF STAGES*	Tr/min RPM			01
REND[T] *EFFICIENCY* (4)	P. ABS BHP RATED	**61.2** **(1)(4)**	kW	01
PUISS. MAX. ABS. ROUE. PROPOSEE MAX. BHP WITH RATED IMPELLER			kW	
H. MANO. MAX. POUR ROUE PROPOSEE *MAX. HEAD WITH RATED IMPELLER*			M.C.L.	
DEBIT CONT. MINI (PAR CONSTR.) *MIN CONTINUOUS FLOW (BY MFR)*			m3/h	
SENS DE ROTATION FACE ACCOUPL. *ROTATION (VIEWED FROM CPLG END)*				

TYPE ET MATERIAUX DE CONSTRUCTION / *CONSTRUCTION AND MATERIALS*

SUPPORTAGE CORPS *CASING MOUNTING*	☐ AXIAL CENTERLINE	■ P. SUR CORPS *FOOT*	☐ P. SUR PALIER *BRACKET*	☐ VERTICAL *VERTICAL*
JOINT *SPLIT*	☐ HORIZONTAL AXIAL	■ VERTICAL RADIAL	☐	
TYPE *TYPE*	■ VOLUTE SIMPLE *SINGLE VOLUTE*	☐ VOLUTE DOUBLE *DOUBLE VOLUTE*	☐ AVEC DIFFUSEUR *DIFFUSER*	
BOSSAGES *TAPPED OPENINGS*	EVENT *VENT*	PURGE *DRAIN*	MANO. *P. GAUGE*	

TUBULURES *NOZZLES*	D.N. *SIZE*	PN OU ASA *PN OR ASA*	FACE DE BRIDE *FACING*	POSITION POSITION
ASPIRATION *SUCTION*	**(3)**	**150 #**	**RF**	**HORIZONTAL**
REFOULEMENT *DISCHARGE*	**(3)**	**150 #**	**RF**	**VERTICAL**

DIAMETER DE LA ROUE *IMPELLER DIAMETER*		MAX *MAX*		TYPE *TYPE*	
PALIER *BEARING*		SOCLE *BASE*			
ACCOUPL[T] MARQUE *COUPLING MFR*		TYPE *TYPE*		MONTE PAR *FITTED BY*	**MFR**
PROTECTEUR D'ACCOUPLEMENT *COUPLING GUARD*	**Anti-sparking**			☐ BOULONS DE FONDATION *BASEPLATE BOLTS*	
☐ PRESSE-ETOUPE (P.E.) PACKING TYPE AND SIZE				AVEC LANTERNE *WITH LANTERN*	
■ GARNITURE MECANIQUE (G.M.) *MECHANICAL SEAL MFR*	**(12)**			TYPE *TYPE*	**Single Seal Plan 11**
G.M. DIAM. ET CARACTERISTIQUES *M.S. SIZE AND CHARACTERISTICS*					
G.M. DESIGNATION SELON API 610 M.S. CLASSIFICATION CODE ACCORDING TO API 610					

EAU DE REFROIDISSEMENT / *COOLING WATER*

SUR DBLE ENVEL. G.M. *TO MECHAN. SEAL JACKET*		m3/h	
SUR DBLE ENVEL. P.E. *TO STUFFING BOX JACKET*		m3/h	
SUR FOND DE POMPE. *TO REAR CASING JACKET*		m3/h	
SUR PALIER *TO BEARING HOUSING*		m3/h	
DEBIT D'EAU TOTAL *TOTAL WATER REQ'D*		m3/h	
EAU LANTERNE P.E. *WATER TO LANTERN RING*			
RECHAUFFAGE HEATING JACKET			
CIRCUIT EAU DE REFROID. *SEAL COOLING*			
SUIVANT API 610 FIG. *ACC. TO API 610 FIG.*	PLAN *PLAN*		

AUXILIAIRES FOURNIS AVEC LA POMPE / *DEVICE SUPPLIED WITH PUMP*

CIRCUIT D'ARROSAGE G.M. *PIPING FOR SEALS*		
SUIVANT API 610 FIG. *ACC. TO API 610 FIG.*	PLAN *PLAN*	01
INSTRUMENT SUR POT G.M. *ALARMS AND SAFETY FOR SEAL DEVICES*		

REMARQUES
REMARKS **General Note : Pump to be insulated for personnel protection.**

(1) VALUE STATED FOR DESIGN FLOWRATE.

(2) DELETED.

(3) TO BE PROVIDED BY VENDOR.

(4) ESTIMATED SHAFT POWER TO BE CONFIRMED BY PUMP VENDOR. MOTOR SHALL BE ENERGY-EFFICIENT : CEI60034-30, CLASS IE2 OR IE3, OR EQUIVALENT NORM.TO BE CONFIRMED BY ELECTRICAL DISCIPLINE IN MECHANICAL DATASHEET (MDS). 01

REFER TO PAGE 3/3 FOR THE OTHER NOTES

POMPE CENTRIFUGE
CENTRI [redacted]

REPERE - *ITEM*

PAGE	DATE	REV
[redacted]	[redacted]	00
[redacted]	[redacted]	01

[redacted]

CLIENT *CUSTOMER*		LIEU D'INSTALLATION *PLANT LOCATION*		REV
FONCTION *SERVICE*	RECYCLED WATER PUMP (10 KG/CM2G)	QUANTITE NECESSAIRE N° OF UNITS	4	
ENTRAINE PAR *DRIVEN BY*	ELECTRIC MOTOR	SERVICE *OPERATION*	CONTINU *CONTINUOUS* (checked) / DISCONTINU *DISCONTINUOUS*	
NOM DU PROJET *PROJECT NAME*		TYPE *TYPE*	Centrifugal Pump	
REFERENCE OFFRE CONSTRUCTEUR *MFR PROPOSAL N°*				

#	MATERIAUX POMPE *MATERIALS PUMP*		GARNITURE MECANIQUE MECHANICAL SEAL		ESAIS CHEZ CONSTRUCTEUR *SHOP TESTS*	DEMANDE *REQUIRED*	CONTR. CLIENT *PURCH. CHECK*
2	ROUE *IMPELLER*	SS 304 (6)	GRAIN FIXE COTE P. *STAT SEAL PUMP SIDE*		PERFORMANCEE *PERFORMANCE*	YES	YES
3	CORPS *INNER CASE*	SS 304 (6)	GRAIN TOURN. COTE P, *ROT. SEAL. PUMP SIDE*		NPSH *NPSH*		
4	CHEMISE (P.E.) *SLEEVE (PACKED)*		SOUFFLET *BELLOWS*	SS	DEMONTAGE *DISMANTLING*		
5	CHEMISE (G.M.) *SLEEVE (M.S.)*	SS 304 (6)	AUTRES PIECES *OTHERS*	SS	PROCES-VERBAL *TEST RECORD*		
6	BAGUE D'USURE *WEAR RINGS*		GRAIN FIXE COTE EXT. *STAT. SEAL EXT. SIDE*		EPREUVE HYDRAU. *HYDRAUL. TESTS*	YES	YES
7	ARBRE *SHAFT*		GRAIN TOURN. C. EXT. *ROT. SEAL EXT. SIDE*		P/T MAX. ADMISSIBLE *P/T MAX. ALLOWABLE*	kg/cm^{2}g	°C
8	JOINTS DE CORPS CASING GASKET		JOINTS *GASKETS & O RINGS*		POIDS POMPE *WEIGHT PUMP* — kg	SOCLE *BASE*	kg
9			POT *RESERVOIR*		LIEU DES ESSAIS *TESTS LOCATION*	At vendor workshop	
0			LIQUIDE TAMPON *BUFFING LIQUID*				

#	MOTEUR FOURNI PAR *MOTOR SUPPLIED BY*		ENTRAINEMENT *DRIVERS*		CARACTERISTIQUES REELLES *MFR FINAL DATA (AS BUILT)*	
2	MARQUE *MFR* / TYPE *TYPE*		ACCOUPLEMENT *COUPLING*		DIAM. REEL ROUE *ACTUAL IMPELLER DIAMETER*	
3	kW *kW* / Tr/min RPM / Carc. Frame	Fte	POULIES/COURROIES *V BELT/PULLEYS*		COUBE D'ESSAI N° *TEST CURVE N°*	
4	PROTECTION *PROTECTION*	IP65			PLAN D'ENSEMBLE N° *OUTLINE DRWG N°*	
5	ANTIDEFLAGRANT *EXPLOSION PROOF*	(13)			PLAN EN COUPE N° *SECTIONAL DRWG N°*	
6	VOLTS/PHASES/Hz *VOLTS/PHASES/Hz*	415/3/50	VARIATION DE VITESSE *VARIABLE SPEED DRIVE UNIT*		PLAN DE LA GARNITURE N° *SEALING SYSTEM DRWG N°*	
7	PALIER *BRGS* / LUB. *LUB.*	Grease			N° DE LA SERIE DE LA POMPE *PUMP SERIAL N°*	
8	INTENSITE A CHARGE MAXI *CURRENT AT MAX LOADING*					
9	MONTE PAR *MOUNTED BY* / POIDS *WT*	kg				

0 REMARQUES
REMARKS

1 (5) DESIGN PRESSURE: SUCTION DESIGN PRESSURE = 0.68 kg/cm2g. DESIGN TEMPERATURE = 98°C.

2 (6) STAINLESS STEEL WITH CARBON CONTENT LESS THAN 0.04%

3 (7) MDMT = 5°C

4 (8) MOTOR RE-ACCELERATION REQUIREMENT TO BE CONSIDERED.

5 (9) NPSHA INDICATED IS FROM PUMP CENTERLINE WHICH IS AT 800 mm ELEVATION FROM GRADE.

6 (10) MOTOR SHALL BE SELECTED FOR END OF THE CURVE.

7 (11) ESTIMATED DISCHARGE DESIGN PRESSURE (CORRESPONDING TO SHUT-OFF PRESSURE) IS 16.5 kg/cm2g. VENDOR SHALL LIMIT RISE TO SHUT-OFF SO AS NOT TO EXCEED THE DISCHARGE DESIGN PRESSURE.

8 (12) SEAL PLAN AND FLUSHING LIQUID REQUIREMENT FOR PUMP TO BE CONFIRMED BY MECHANICAL DISCIPLINE IN MDS BASED ON VENDOR REQUIREMENT.

9 (13) MOTOR TYPE BASED ON HAZARDOUS AREA CLASSIFICATION TO BE CONFIRMED BY ELECTRICAL AND HSE DISCIPLINE IN MDS.

0

POMPE CENTRIFUGE
CENTRI [redacted]

REPERE - *ITEM*

PAGE | DATE | REV
00
01